

ENV 6106

Air Pollution Modeling

Spring 2023

Topics

- AERSCREEN demo
- Expectation
 - Be able to run AERSCREEN and familiar with input parameters at least for point source
- Readings: AERSCREEN user guide
 - https://gaftp.epa.gov/Air/aqmg/SCRAM/models/screening/aerscreen/aerscreen_userguide.pdf

AERSCREEN

- An easy-to-use, interactive air quality screening model
- The purpose is to quickly and easily screen a source to determine the worst case or maximum concentrations that might occur
- Do screening is to eliminate the need to run a refined, and more costly, air quality model and analysis
- “AERSCREEN View” is a graphical interface to AERSCREEN developed by Lake Environment
 - AERSCREEN is open source and free. “AERSCREEN View” is not

AERSCREEN

- Based on only a few inputs, AERSCREEN automatically configures and runs AERMOD
 - Eliminated much efforts in running a detailed AERMOD analysis
- AERSCREEN download webpage:
 - <https://www.epa.gov/scram/air-quality-dispersion-modeling-screening-models>
 - Download these
 - AERSCREEN & MAKEMET code and executables
 - AERSCREEN user guide
 - Testcase & README txt file

AERSCREEN components

- AERSCREEN calls these components when needed
 - MAKEMET: generate a combinations of meteorological conditions for AERSCREEN
 - AERMOD: the actual dispersion model AERSCREEN is running, but with all parameters auto configured based on user inputs
 - BPIPPRM: building downwash module, used only when downwash is enabled
 - AERSURFACE: generate surface characteristics, used only when detailed surface characteristics are needed
 - AERMAP: generate ground elevation data, used only when detailed elevation data are needed
- Only needs to run AERSCREEN
 - Input data may need to be prepared for each component

Initial information

- AERSCREEN will attempt to read “aerscreen.inp”, which was generated during previous runs and contains all information needed
 - If no such file, then start a new run
- Source type:
 - POINT: regular point source (a stack)
 - POINTCAP: stack with a rain hat
 - POINTHOR: horizontal stack
 - FLARE: model flare burning
 - VOLUME: volume source (e.g. a cube emitting from all sides)
 - AREA: rectangle source (e.g. land fill)
 - AREACIRC: Circular area sources

Source information

- Information on emissions from source
 - Different for different source types
 - We will focus on point source only
 - Emission Rate (g/s), Stack Height (meters): Stack Diameter (meters), Stack Temperature (K)
 - Exit Velocity or Flow Rate
- Others
 - Rural or urban: affect how vertical wind varies, and surface characteristics
 - Minimum ambient distance: closest receptor distance downwind of source
 - NO_2 chemistry: only for NO_2 , calculate ratios of NO and NO_2

Building downwash

- Option to use an existing BPIPPRM input file
 - Bypass following inputs, details see addendum:
 - <https://www.epa.gov/scram/air-quality-dispersion-modeling-related-model-support-programs#bpipprm>
- Building height (feet or meters)
- Maximum building horizontal dimension (feet or meters)
- Minimum building horizontal dimension (feet or meters)
- Degrees from North of maximum building horizontal dimension (0-179 degrees)
- Degrees from North of stack location relative to building center (0-360 degrees)
- Distance between stack and building center (feet or meters)

Building downwash

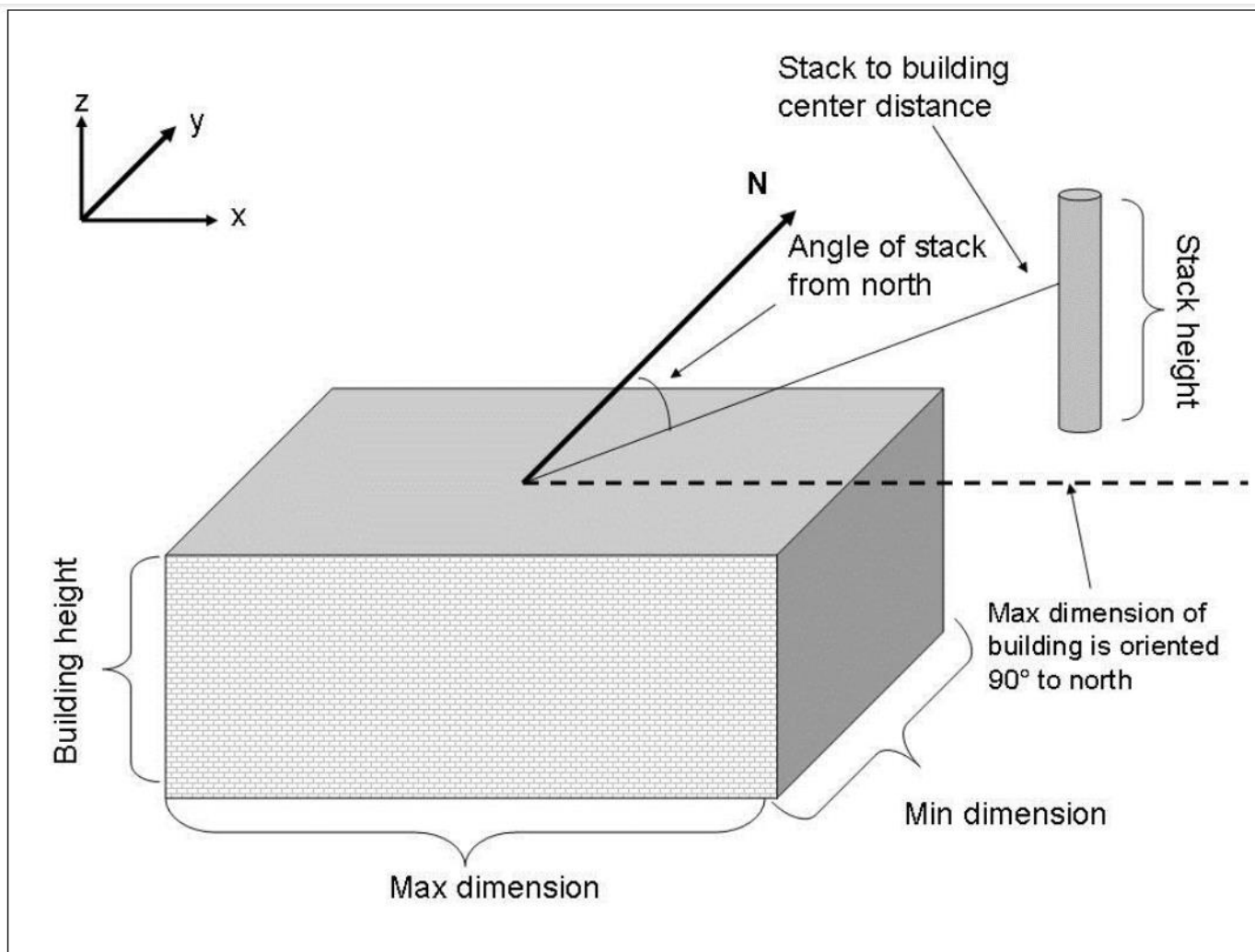
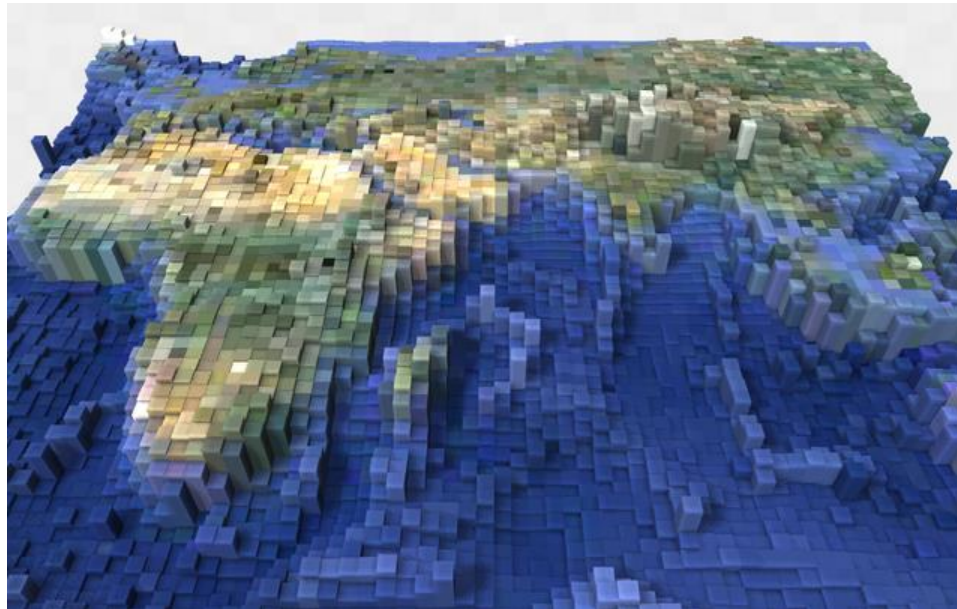


Figure 11. Stack and building orientation for a building oriented 90 degrees to north and stack oriented 45 degrees to north.

Terrain

- If choose to account for spatial varying terrain variation, need to have external terrain elevation data
 - Data file name listed in demlist.txt
 - We will discuss more when we come to AERMAP



An example of digital elevation map

<https://gis.stackexchange.com/questions/68965/precision-vs-resolution-in-elevation-data>

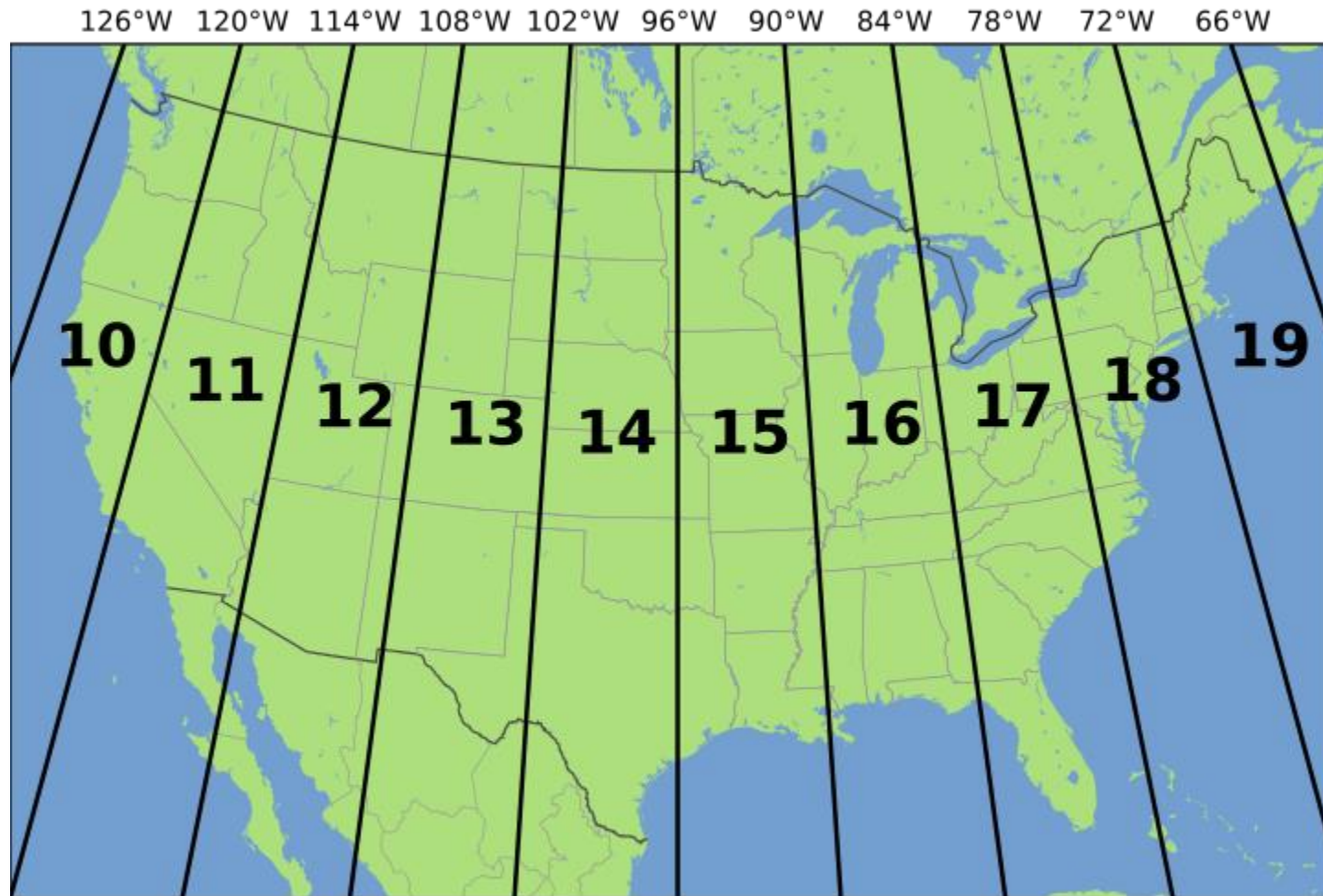
Terrain

- Probe distance (meters)
 - Furthest distance to calculate concentration
- Include discrete receptor distances
 - Option to include user defined location for conc calc
- Flagpole receptors
 - Calculate concentration at higher elevation
- Source elevation, or AERMAP determine source elevation from digital terrain data
- Source coordinates (geographic or UTM)
- NAD datum (NAD 27 or 83)
- UTM zone (if UTM coordinates entered)

NAD & UTM

- Used to account for the curvature of earth
- NAD: North American Datum of 1927/1983
 - Datum: coordinate system defines the shape of earth
 - NAD 27/83 best suits north American
- UTM: Universal Transverse Mercator coordinate system
 - A simple algorithm to expand round earth to a flat surface, using a datum
 - Instead of uneven lat and lon, now you have a evenly distributed X and Y

UTM zone in US



World Mercator projection with country going to true size



@neilrkaye

<https://i.redd.it/96ptbqzyfr11.gif>

Meteorology data

- Minimum and maximum ambient air temperatures (Fahrenheit or Kelvin)
- Minimum wind speed (m/s)
 - Recall dispersion model do not work well under calm condition
- Anemometer height (m)
 - 10 m is the standard height
- Surface characteristics
 - Can be user-entered single value; AERMET tables or listed in an external file (varying by certain time periods)
 - We will discuss the time varying option in AERMET

Meteorology data

- Albedo: % of incoming radiation reflected back
 - 0 for perfect black body, 1 for perfect reflective surface
- Bowen ratio: ratio between sensible heat and latent heat
 - Sensible heat: energy used to increase temp
 - Latent heat: energy used to change phase, no temp change
 - How much energy were used to evaporate water, but not used to increase temperature
 - < 0.1 for large water body and up to 10 for dry dessert
- Surface roughness length (Z_0)
 - Lowest height at which wind speed is zero
 - Over Z_0 wind increases according to a power law
 - < 1 mm in very flat surface, a few meter for urban region

Meteorology data

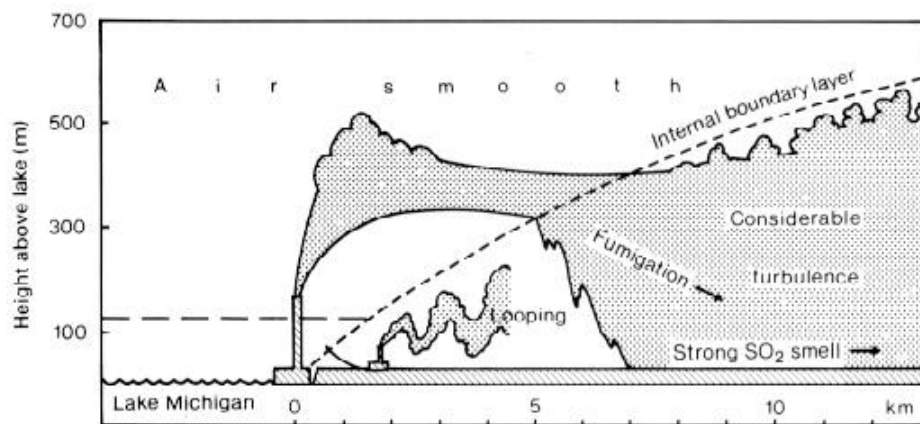
- Surface friction velocity (u^*)
 - A scaling parameter. a measure of the vertical transport of horizontal momentum
 - Past studies showed that under low wind speed and stable condition, u^* need to be modified to get better results
 - Optional to adjust u^*
- Convective velocity (w^*)
 - Internally calculated, describes the speed of vertical transport due to convection (buoyancy gained from temp difference)
- Monin-Obukhov length (L)
 - Internally calculated, one way to describes stability
 - $L < 0$: unstable, $L > 0$: stable

Meteorology data

- A look up table for Albedo, Bowen ratio and Z_0 (from CARB) is provided on webcourse
- AERMOD/AERSCREEN uses two meteorological files
 - Surface observation and vertical distribution profile
 - Default: .sfc and .pfl
 - Collected at surface met stations and usually from balloon sounding (aka upper air sounding)
 - <https://www.youtube.com/watch?v=9v-aqz18pul>

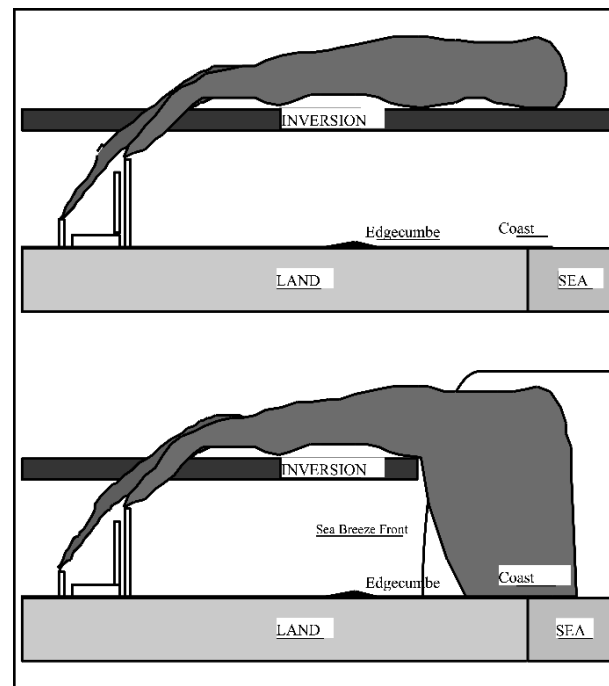
Fumigation

- AERSCREEN can account for two fumigations
 - Inversion break-up and shoreline
 - Inherited from SCREEN3 model
 - Only for certain conditions
 - Elevated inversion



Shoreline fumigation

https://atmos.washington.edu/~justin/radar_project/clearair.htm



Inversion break-up fumigation

https://www.researchgate.net/figure/Figure-9-Illustrating-the-effect-of-inversion-break-up_283653069_fig4

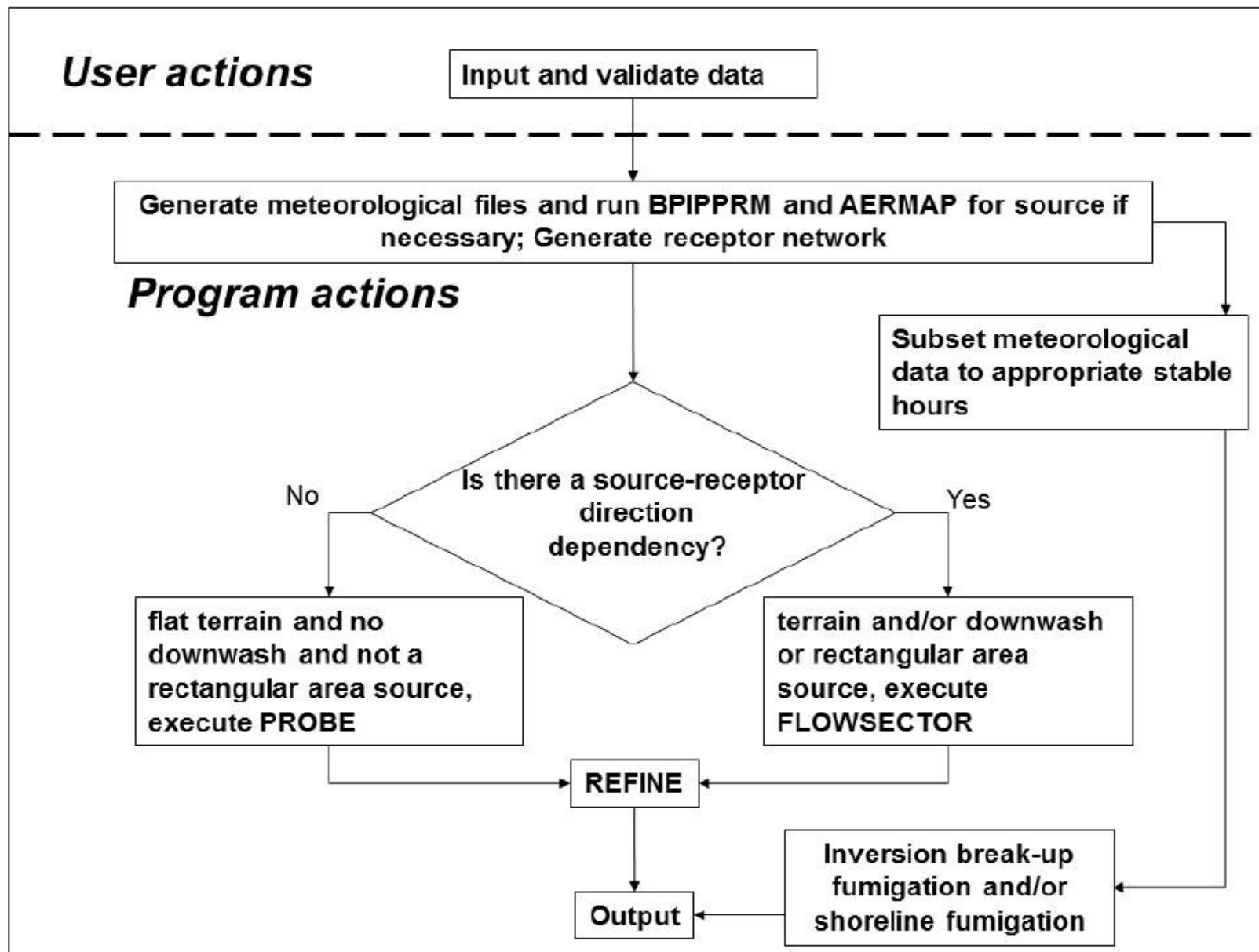


Figure 21. AERSCREEN processing and stages.

Fig. 21. AERSCREEN flowchart

Important AERSCREEN outputs

- Output data: *user_defined_name.out*
 - Summary of input parameters and output concentration data
- Log file: *user_defined_name.log*
 - Logs from AERSCREEN run
- Restart file: *aerscreen.inp*, *user_defined_name.inp*
 - Stores all inputs information and can serve as a template for a quick next-round run